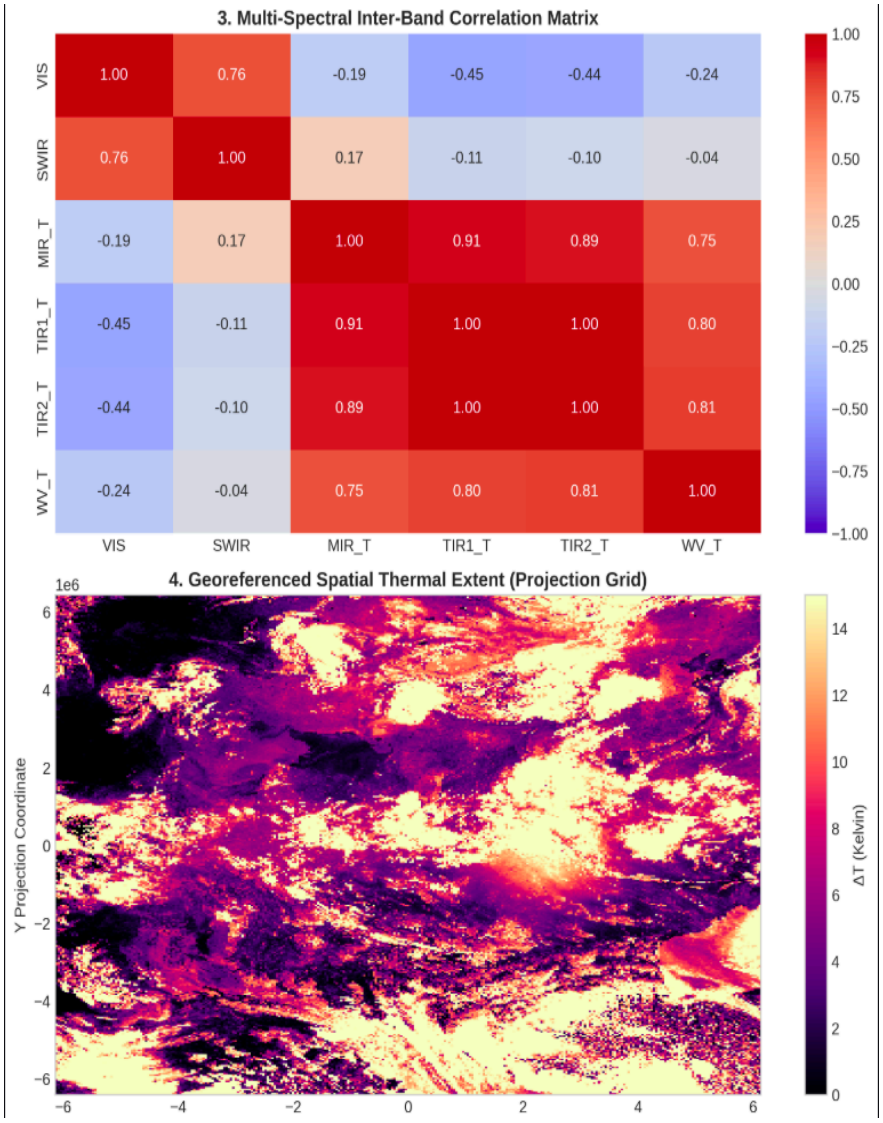
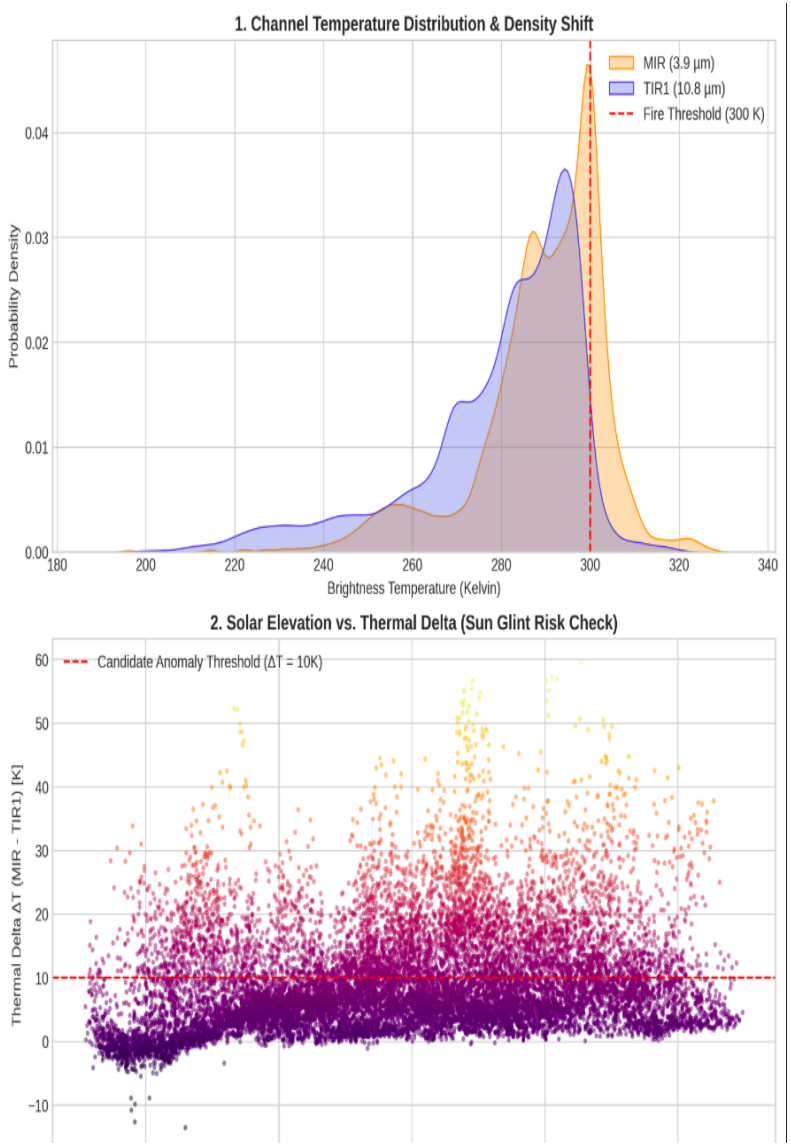
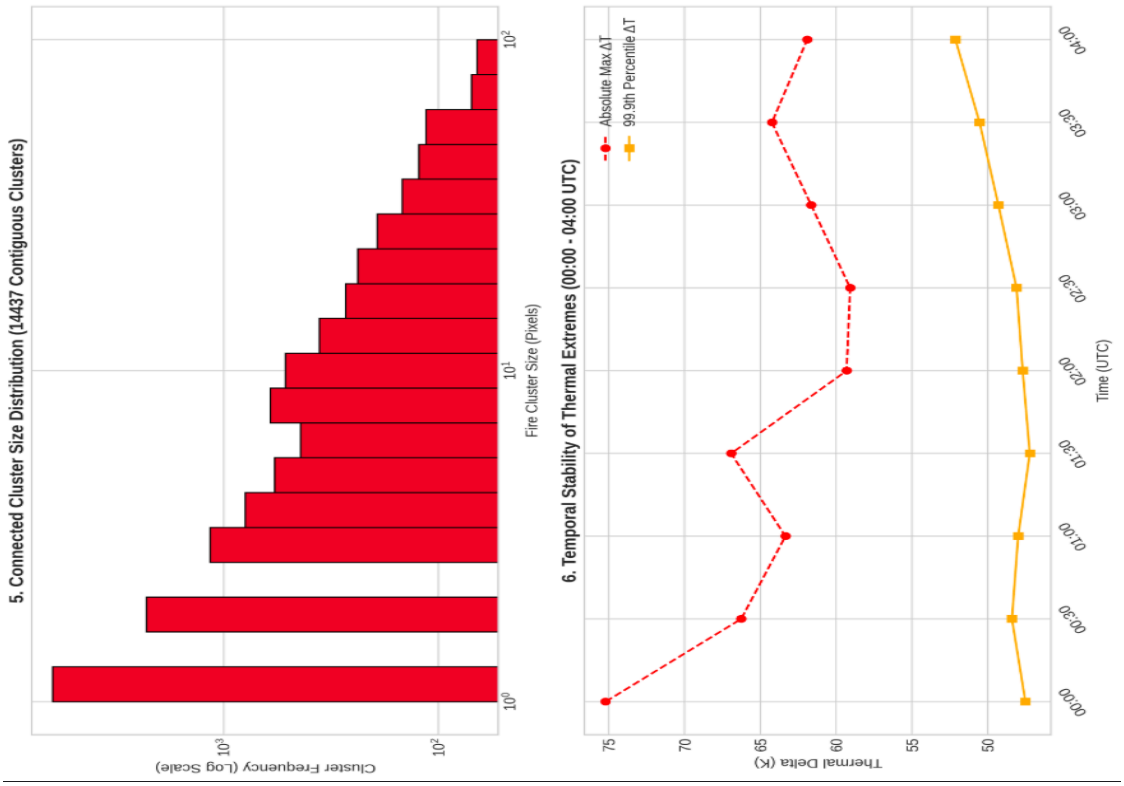
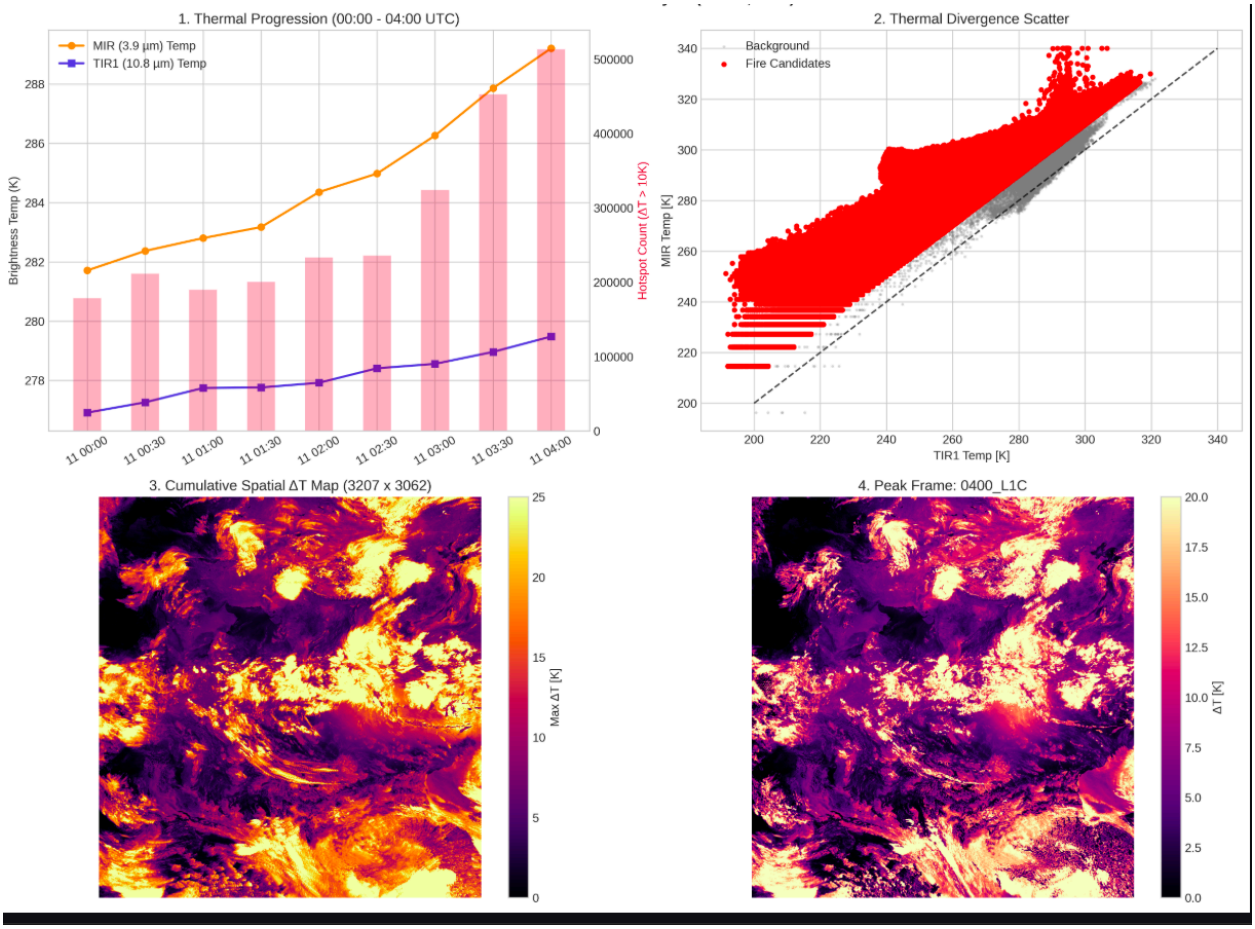

3. Multi-Spectral Inter-Band Correlation Matrix
VIS
SWIR
MIR_T
TIR1_T
TIR2_T
WV_T
1.00 0.76 -0.19 -0.45 -0.44 -0.24
0.76 1.00 0.17 -0.11 -0.10 -0.04
-0.19 0.17 1.00 0.91 0.89 0.75
-0.45 -0.11 0.91 1.00 1.00 0.80
-0.44 -0.10 0.89 1.00 1.00 0.81
-0.24 -0.04 0.75 0.80 0.81 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
4. Georeferenced Spatial Thermal Extent (Projection Grid)
1e6
Y Projection Coordinate
6
4
2
0
−2
−4
−6
−6 −4 −2 0 2 4 6
ΔT (Kelvin)
14
12
10
8
6
4
2
0


1. Channel Temperature Distribution & Density Shift
MIR (3.9 μm)
TIR1 (10.8 μm)
Fire Threshold (300 K)
Probability Density
0.04
0.03
0.02
0.01
0.00
180 200 220 240 260 280 300 320 340
Brightness Temperature (Kelvin)
2. Solar Elevation vs. Thermal Delta (Sun Glint Risk Check)
Candidate Anomaly Threshold (ΔT = 10K)
Thermal Delta ΔT (MIR - TIR1) [K]
60
50
40
30
20
10
0
−10


5. Connected Cluster Size Distribution (14437 Contiguous Clusters)
Cluster Frequency (Log Scale)
Fire Cluster Size (Pixels)
6. Temporal Stability of Thermal Extremes (00:00 - 04:00 UTC)
Absolute Max ΔT
99.9th Percentile ΔT
Thermal Delta (K)
Time (UTC)


1. Thermal Progression (00:00 - 04:00 UTC)
MIR (3.9 µm) Temp
TIR1 (10.8 µm) Temp
Brightness Temp (K)
Hotspot Count (ΔT > 10K)
2. Thermal Divergence Scatter
Background
Fire Candidates
MIR Temp [K]
TIR1 Temp [K]
3. Cumulative Spatial ΔT Map (3207 x 3062)
Max ΔT [K]
4. Peak Frame: 0400_L1C
ΔT [K]